

CJC 2021 H2 Biology (Syllabus 9744) Preliminary Examination Paper 4

Apparatus and Materials List for Question 1

	Apparatus/ Reagents/ Chemicals	Quantity per candidate
1	D	3 cm ³ (in plastic container)
2	L (on ice in Styrofoam cup)	10 cm ³ (in plastic container)
3	Pasteur pipettes	1 (in plastic bag)
4	1 cm ³ syringes	6 (in plastic bag)
5	6 cm x 15 cm filters, folded longitudinally to form a tent, of the following colours: (i) U - purple (ii) G - green (iii) O - orange (iv) R - red	1 of each colour (in basket)
6	White tile	1 (in basket)
7	Long piece of aluminum foil	1 (in plastic bag)
8	Square piece of aluminum foil	1 (in plastic bag)
9	Bench lamp	1
10	Stopwatch	1 (in basket)
11	30 cm ruler	1 (in basket)
12	Glass rod	1 (in basket)
13	Safety goggles	1 (in basket)
14	Container labelled for washing	1 (in basket)
15	Container labelled for waste	1 (in basket)
16	Paper towels	3 (in plastic bag)
17	Test tubes	7 (in plastic bag)
18	Test tube rack	1
19	5 cm ³ syringes	2 (in plastic bag)
20	0.1% DCPIP	15 cm ³ (in plastic container)
21	1% ascorbic acid	30 cm ³ (in plastic container)
22	J	10 cm ³ (in plastic container)
23	W	20cm ³ (in plastic container)

Apparatus and Materials List for Question 2

	Apparatus/ Reagents/ Chemicals	Quantity per candidate
1	Microscope (fitted with eyepiece graticule)	1 (half duration)
2	30 cm plastic ruler (also for Question 1)	1
3	Prepared Slide, labelled E1 (GCE A Level 1995)	1 (half duration)

Preparation List for Chemicals for Question 1:

1% Ascorbic Acid: 0.2% (actual concentration)

J: 0.1% ascorbic acid

0.1% DCPIP: 0.1g DCPIP in 100ml DI H₂O

D: 0.2g DCPIP + 0.9g KCl in 250ml Phosphate buffer

L: 15g of spinach leaves for every 100ml of cold extraction buffer

W: Distilled water